Untitled

2026-02-24

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
df <- read_csv("/Users/xiaoyuzhou/Desktop/Doodling Result.csv")
```

```
## New names:
## Rows: 54 Columns: 167
## -- Column specification
## -------------------------------------------------------- Delimiter: "," chr
## (7): p_id, auditory_condition, visual_condition, sexual_orientation_4_... dbl
## (158): group_number, visual_1, visual_2, visual_3, visual_4, visual_5, v... num
## (1): diagnostic lgl (1): gender_4_TEXT
## i Use `spec()` to retrieve the full column specification for this data. i
## Specify the column types or set `show_col_types = FALSE` to quiet this message.
## * `` -> `...166`
## * `` -> `...167`
```

```r
answer_key <- read_excel("/Users/xiaoyuzhou/Desktop/recall_answer_key.xlsx")

head(df)
```

```
## # A tibble: 6 x 167
##   p_id  group_number auditory_condition visual_condition visual_1 visual_2
##   <chr>        <dbl> <chr>              <chr>                <dbl>    <dbl>
## 1 1                1 D                  A                        1        1
## 2 2                2 B                  A                        1        1
## 3 3                1 A                  A                        1        1
## 4 4                2 C                  A                        2        1
## 5 5                1 C                  D                        1        1
## 6 6                2 A                  A                        1        1
## # i 161 more variables: visual_3 <dbl>, visual_4 <dbl>, visual_5 <dbl>,
## #   visual_6 <dbl>, visual_7 <dbl>, visual_8 <dbl>, visual_9 <dbl>,
## #   visual_10 <dbl>, visual_11 <dbl>, visual_12 <dbl>, gender <dbl>,
## #   gender_4_TEXT <lgl>, sexual_orientation <dbl>,
## #   sexual_orientation_4_TEXT <chr>, age <dbl>, diagnostic <dbl>,
## #   diagnostic_14_TEXT <chr>, gpa <dbl>, race <dbl>, race_10_TEXT <chr>,
## #   learn_type_1 <dbl>, learn_type_2 <dbl>, learn_type_3 <dbl>, ...
```

```r
print(df)
```

```
## # A tibble: 54 x 167
##    p_id  group_number auditory_condition visual_condition visual_1 visual_2
##    <chr>        <dbl> <chr>              <chr>                <dbl>    <dbl>
##  1 1                1 D                  A                        1        1
##  2 2                2 B                  A                        1        1
##  3 3                1 A                  A                        1        1
##  4 4                2 C                  A                        2        1
##  5 5                1 C                  D                        1        1
##  6 6                2 A                  A                        1        1
##  7 7                1 C                  A                        1        2
##  8 8                2 C                  A                        1        1
##  9 9                1 A                  A                        1        1
## 10 11               1 B                  B                        1        1
## # i 44 more rows
## # i 161 more variables: visual_3 <dbl>, visual_4 <dbl>, visual_5 <dbl>,
## #   visual_6 <dbl>, visual_7 <dbl>, visual_8 <dbl>, visual_9 <dbl>,
## #   visual_10 <dbl>, visual_11 <dbl>, visual_12 <dbl>, gender <dbl>,
## #   gender_4_TEXT <lgl>, sexual_orientation <dbl>,
## #   sexual_orientation_4_TEXT <chr>, age <dbl>, diagnostic <dbl>,
## #   diagnostic_14_TEXT <chr>, gpa <dbl>, race <dbl>, race_10_TEXT <chr>, ...
```

```r
colnames(df)
```

```
##   [1] "p_id"                  "group_number"
##   [3] "auditory_condition"    "visual_condition"
##   [5] "visual_1"              "visual_2"
##   [7] "visual_3"              "visual_4"
##   [9] "visual_5"              "visual_6"
##  [11] "visual_7"              "visual_8"
```

```
##  [13] "visual_9"                "visual_10"
##  [15] "visual_11"               "visual_12"
##  [17] "gender"                  "gender_4_TEXT"
##  [19] "sexual_orientation"      "sexual_orientation_4_TEXT"
##  [21] "age"                     "diagnostic"
##  [23] "diagnostic_14_TEXT"      "gpa"
##  [25] "race"                    "race_10_TEXT"
##  [27] "learn_type_1"            "learn_type_2"
##  [29] "learn_type_3"            "learn_type_4"
##  [31] "learn_type_5"            "learn_type_6"
##  [33] "learn_type_7"            "learn_type_8"
##  [35] "learn_type_9"            "learn_type_10"
##  [37] "learn_type_11"           "learn_type_12"
##  [39] "learn_type_13"           "learn_type_14"
##  [41] "learn_type_15"           "learn_type_16"
##  [43] "learn_type_17"           "learn_type_18"
##  [45] "learn_type_19"           "learn_type_20"
##  [47] "learn_type_21"           "learn_type_22"
##  [49] "learn_type_23"           "learn_type_24"
##  [51] "learn_type_25"           "learn_type_26"
##  [53] "learn_type_27"           "learn_type_28"
##  [55] "learn_type_29"           "learn_type_30"
##  [57] "learn_type_31"           "learn_type_32"
##  [59] "learn_type_33"           "learn_type_34"
##  [61] "learn_type_35"           "learn_type_36"
##  [63] "learn_type_37"           "learn_type_38"
##  [65] "learn_type_39"           "learn_type_40"
##  [67] "learn_type_41"           "learn_type_42"
##  [69] "learn_type_43"           "learn_type_44"
##  [71] "BAI_1"                   "BAI_2"
##  [73] "BAI_3"                   "BAI_4"
##  [75] "BAI_5"                   "BAI_6"
##  [77] "BAI_7"                   "BAI_8"
##  [79] "BAI_9"                   "BAI_10"
##  [81] "BAI_11"                  "BAI_12"
##  [83] "BAI_13"                  "BAI_14"
##  [85] "BAI_15"                  "BAI_16"
##  [87] "BAI_17"                  "BAI_18"
##  [89] "BAI_19"                  "BAI_20"
##  [91] "BAI_21"                  "MPI_1"
##  [93] "MPI_2"                   "MPI_3"
##  [95] "MPI_4"                   "MPI_5"
##  [97] "MPI_6"                   "MPI_7"
##  [99] "MPI_8"                   "MPI_9"
## [101] "MPI_10"                  "MPI_11"
## [103] "MPI_12"                  "MPI_13"
## [105] "MPI_14"                  "big_five_1"
## [107] "big_five_2"              "big_five_3"
## [109] "big_five_4"              "big_five_5"
## [111] "big_five_6"              "big_five_7"
## [113] "big_five_8"              "big_five_9"
## [115] "big_five_10"             "big_five_11"
## [117] "big_five_12"             "big_five_13"
## [119] "big_five_14"             "big_five_15"
```

```
## [121] "big_five_16"               "big_five_17"
## [123] "big_five_18"               "big_five_19"
## [125] "big_five_20"               "big_five_21"
## [127] "big_five_22"               "big_five_23"
## [129] "big_five_24"               "big_five_25"
## [131] "big_five_26"               "big_five_27"
## [133] "big_five_28"               "big_five_29"
## [135] "big_five_30"               "big_five_31"
## [137] "big_five_32"               "big_five_33"
## [139] "big_five_34"               "big_five_35"
## [141] "big_five_36"               "big_five_37"
## [143] "big_five_38"               "big_five_39"
## [145] "big_five_40"               "big_five_41"
## [147] "big_five_42"               "big_five_43"
## [149] "big_five_44"               "audio_1"
## [151] "audio_2"                   "audio_3"
## [153] "audio_4"                   "audio_5"
## [155] "audio_6"                   "audio_7"
## [157] "audio_8"                   "audio_9"
## [159] "audio_10"                  "audio_11"
## [161] "audio_12"                  "self_report_1"
## [163] "self_report_2"             "self_report_3"
## [165] "self_report_4"             "...166"
## [167] "...167"
```

```r
# pivot longer

long_df <- df %>%
  pivot_longer(
    cols = matches("^(visual|audio)_\\d+$"),
    names_to = "item",
    values_to = "response"
  ) %>%
  mutate(
    TaskType = if_else(str_starts(item, "visual_"), "Visual", "Auditory"),
    QuestionNum = as.integer(str_extract(item, "\\d+")),
    PassageIndex = ceiling(QuestionNum / 4),  # 1-4 => passage 1; 5-8 => passage 2; 9-12 => passage 3
    ConditionGroup = if_else(TaskType == "Visual", visual_condition, auditory_condition),
    response = as.integer(response) # 1 = yes, 2 = no
  )

# add the passage title
long_df <- long_df %>%
  mutate(
    PassageName = case_when(
      TaskType == "Visual"   & PassageIndex == 1 ~ "macOS vs. Linux vs. Windows",
      TaskType == "Visual"   & PassageIndex == 2 ~ "Bash vs. Zsh",
      TaskType == "Visual"   & PassageIndex == 3 ~ "File systems",
      TaskType == "Auditory" & PassageIndex == 1 ~ "Package Managers",
      TaskType == "Auditory" & PassageIndex == 2 ~ "Security models",
      TaskType == "Auditory" & PassageIndex == 3 ~ "GUI vs. CLI",
      TRUE ~ NA_character_
    )
  )
```

```r
# add doodling type
lookup <- tribble(
  ~TaskType,   ~ConditionGroup, ~PassageIndex, ~PassageName,                          ~doodle_type,

  # Auditory
  "Auditory",  "A",             1,             "Package Managers",                  "Free",
  "Auditory",  "A",             2,             "Security models",                   "Structured",
  "Auditory",  "A",             3,             "GUI vs. CLI",                       "Control",

  "Auditory",  "B",             1,             "Package Managers",                  "Control",
  "Auditory",  "B",             2,             "Security models",                   "Free",
  "Auditory",  "B",             3,             "GUI vs. CLI",                       "Structured",

  "Auditory",  "C",             1,             "Package Managers",                  "Structured",
  "Auditory",  "C",             2,             "Security models",                   "Control",
  "Auditory",  "C",             3,             "GUI vs. CLI",                       "Free",

  "Auditory",  "D",             1,             "Pcakage Managers",                  "Free",
  "Auditory",  "D",             2,             "Security models",                   "Control",
  "Auditory",  "D",             3,             "GUI vs. CLI",                       "Structured",

  # Visual
  "Visual",    "A",             1,             "macOS vs. Linux vs. Windows",       "Free",
  "Visual",    "A",             2,             "Bash vs. Zsh",                      "Structured",
  "Visual",    "A",             3,             "File systems",                      "Control",

  "Visual",    "B",             1,             "macOS vs. Linux vs. Windows",       "Control",
  "Visual",    "B",             2,             "Bash vs. Zsh",                      "Free",
  "Visual",    "B",             3,             "File systems",                      "Structured",

  "Visual",    "C",             1,             "macOS vs. Linux vs. Windows",       "Structured",
  "Visual",    "C",             2,             "Bash vs. Zsh",                      "Control",
  "Visual",    "C",             3,             "File systems",                      "Free",

  "Visual",    "D",             1,             "macOS vs. Linux vs. Window",        "Control",
  "Visual",    "D",             2,             "File systems",                      "Free",
  "Visual",    "D",             3,             "Bash vs. Zsh",                      "Structured"
)


# join the datasets

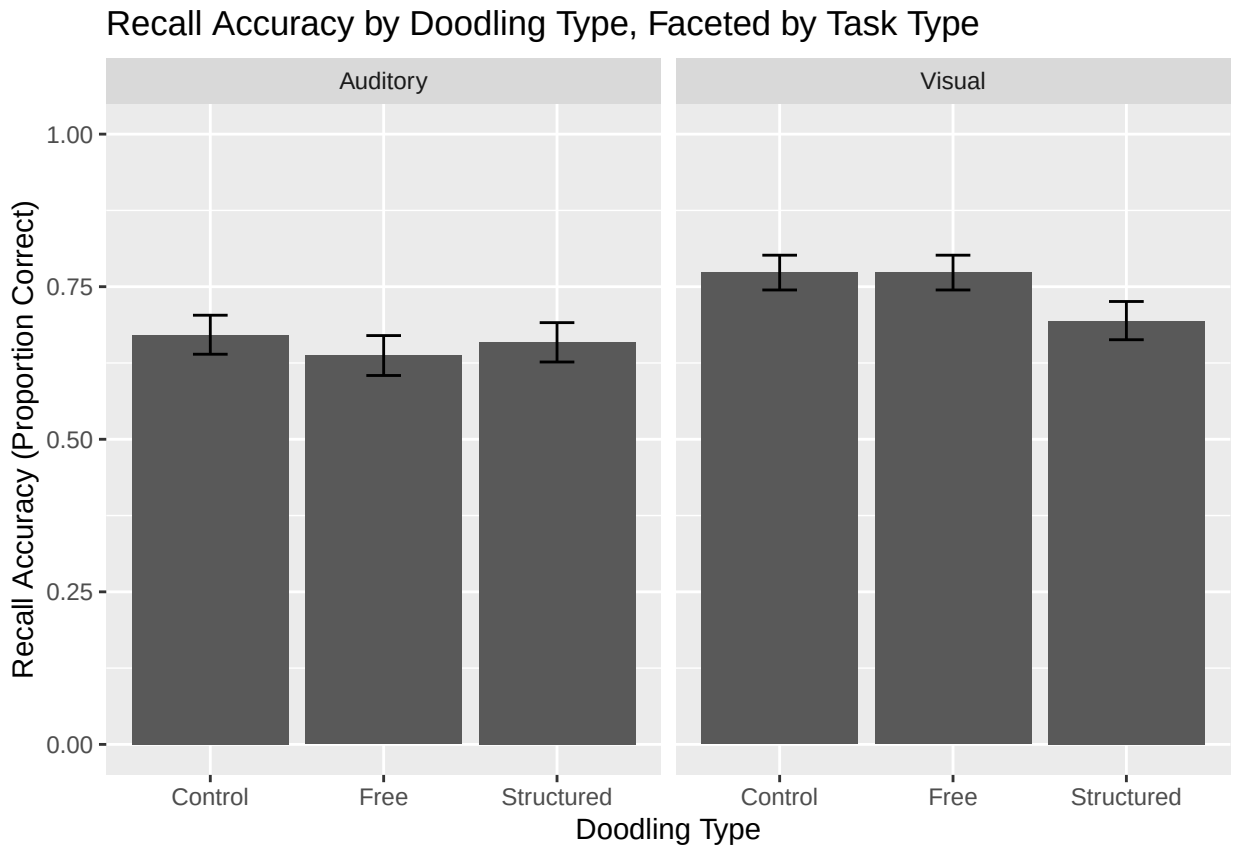
long_df <- long_df %>%
  left_join(lookup, by = c("TaskType", "ConditionGroup","PassageIndex")) %>%
  mutate(
    doodle_type = factor(doodle_type, levels = c("Control","Free","Structured")),
    TaskType = factor(TaskType),
    p_id = factor(p_id)
  )

# join the answer key and compute accuracy
long_df <- long_df %>%
```

```
  left_join(answer_key, by = "item") %>%
  mutate(
    RecallAcc = as.integer(response == correct_answer)
  )
```

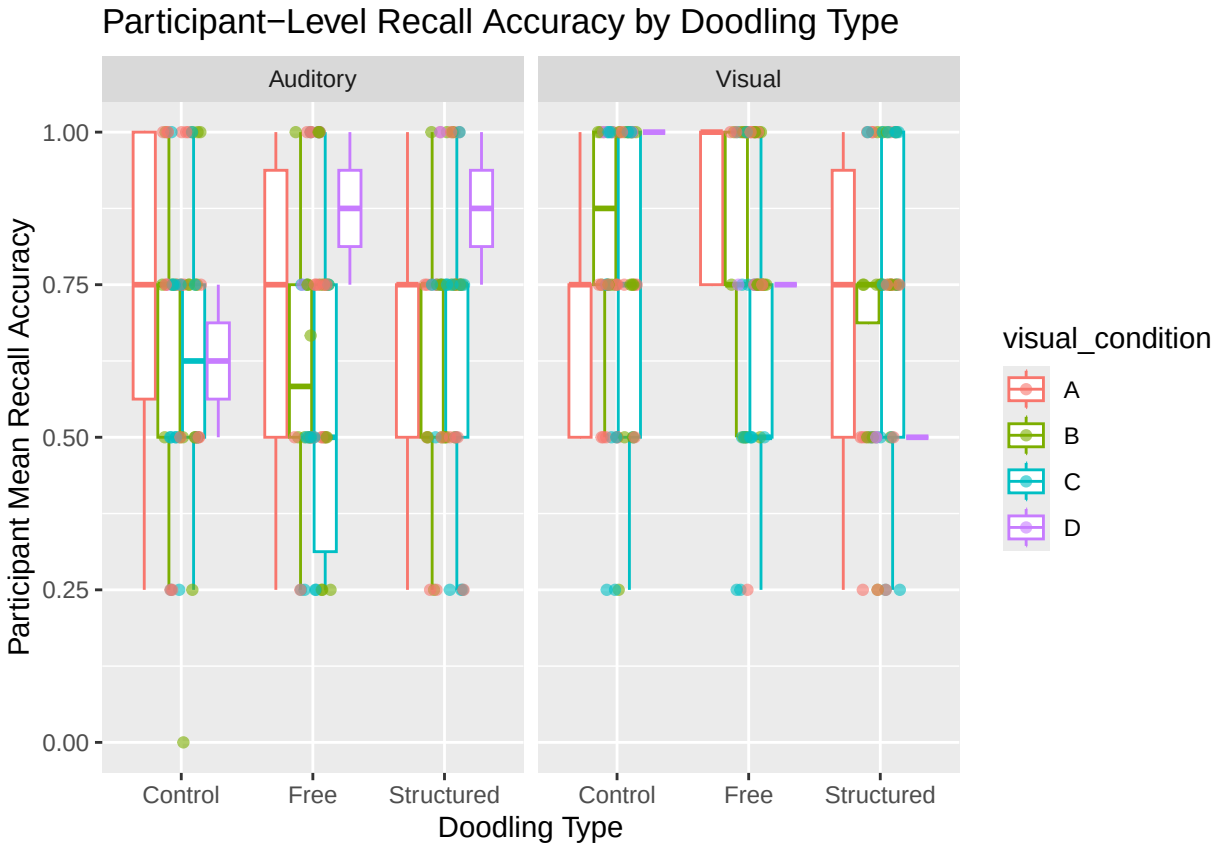
```
# 1) Summarize accuracy by TaskType × doodle_type
plot_df <- long_df %>%
  group_by(TaskType, doodle_type) %>%
  summarise(
    n = n(),
    acc = mean(RecallAcc, na.rm = TRUE),
    se = sqrt(acc * (1 - acc) / n),   # binomial SE for proportions
    .groups = "drop"
  )

# 2) Plot
ggplot(plot_df, aes(x = doodle_type, y = acc)) +
  geom_col() +
  geom_errorbar(aes(ymin = acc - se, ymax = acc + se), width = 0.2) +
  facet_wrap(~ TaskType) +
  labs(
    x = "Doodling Type",
    y = "Recall Accuracy (Proportion Correct)",
    title = "Recall Accuracy by Doodling Type, Faceted by Task Type"
  ) +
  coord_cartesian(ylim = c(0, 1))
```




```r
pp_df <- long_df %>%
  group_by(p_id, TaskType, doodle_type, visual_condition) %>%
  summarise(acc = mean(RecallAcc, na.rm = TRUE), .groups = "drop")

ggplot(pp_df, aes(x = doodle_type, y = acc, color = visual_condition)) +
  geom_boxplot(outlier.shape = NA) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.6) +
  facet_wrap(~ TaskType) +
  labs(
    x = "Doodling Type",
    y = "Participant Mean Recall Accuracy",
    title = "Participant-Level Recall Accuracy by Doodling Type"
  ) +
  coord_cartesian(ylim = c(0, 1))
```




```
pp_df %>%
  group_by(TaskType, doodle_type) %>%
  summarise(sd_acc = sd(acc))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by TaskType and doodle_type.
## i Output is grouped by TaskType.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(TaskType, doodle_type))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.

## # A tibble: 6 x 3
## # Groups:   TaskType [2]
##   TaskType doodle_type sd_acc
##   <fct>    <fct>        <dbl>
## 1 Auditory Control      0.247
## 2 Auditory Free         0.246
## 3 Auditory Structured   0.229
## 4 Visual   Control      0.229
## 5 Visual   Free         0.229
## 6 Visual   Structured   0.246
```

```
pp_df %>%
  group_by(TaskType, doodle_type) %>%
  summarise(mean_acc = mean(acc))
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by TaskType and doodle_type.
## i Output is grouped by TaskType.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(TaskType, doodle_type))` for per-operation grouping
##   (`?dplyr::dplyr_by`) instead.
```

```
## # A tibble: 6 x 3
## # Groups:   TaskType [2]
##   TaskType doodle_type mean_acc
##   <fct>    <fct>          <dbl>
## 1 Auditory Control        0.671
## 2 Auditory Free           0.637
## 3 Auditory Structured     0.662
## 4 Visual   Control        0.773
## 5 Visual   Free           0.773
## 6 Visual   Structured     0.694
```

- as the participants number increases the mean_acc increases and the sd decreases.

```
model1 <- glmer(RecallAcc ~ TaskType * doodle_type + (1 | p_id),
      family = binomial,
      data = long_df)

summary(model1)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: RecallAcc ~ TaskType * doodle_type + (1 | p_id)
##    Data: long_df
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1566.3    1602.4    -776.1    1552.3      1286
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2730 -1.2597  0.5725  0.6791  0.9940
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  p_id   (Intercept) 0.1158   0.3403
## Number of obs: 1293, groups:  p_id, 54
##
## Fixed effects:
##                                    Estimate Std. Error z value Pr(>|z|)
## (Intercept)                         0.73361    0.15400   4.764  1.9e-06 ***
## TaskTypeVisual                      0.52390    0.21993   2.382   0.0172 *
## doodle_typeFree                    -0.15504    0.20521  -0.756   0.4499
## doodle_typeStructured              -0.05784    0.20684  -0.280   0.7798
## TaskTypeVisual:doodle_typeFree      0.15504    0.30956   0.501   0.6165
## TaskTypeVisual:doodle_typeStructured -0.35646    0.30324  -1.176   0.2398
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) TskTyV ddl_tF ddl_tS TTV:_F
## TaskTypeVsl -0.632
## doodl_typFr -0.680  0.476
## ddl_typStrc -0.675  0.472  0.506
## TskTypVs:_F  0.451 -0.710 -0.663 -0.336
## TskTypVs:_S  0.459 -0.725 -0.345 -0.682  0.515
```

Learner's Type

```
visual_verbal_items <- c(
   "learn_type_3", "learn_type_7", "learn_type_11", "learn_type_15",
  "learn_type_19", "learn_type_23", "learn_type_27", "learn_type_31",
  "learn_type_35", "learn_type_39", "learn_type_43")

# change the value
df <- df %>%
  mutate(
    across(
      all_of(visual_verbal_items),
      ~ case_when(
        . == 1 ~ -1,   # 1 = Visual
        . == 2 ~  1,   # 2 = Verbal
        TRUE ~ NA_real_
      ),
      .names = "{.col}_scored"
    )
  ) %>%
  rowwise() %>%
  mutate(
    visual_verbal_score = sum(c_across(ends_with("_scored")), na.rm = FALSE)
  ) %>%
  ungroup()

long_df <- long_df %>%
  mutate(p_id = as.character(p_id)) %>%
  left_join(
    df %>%
      mutate(p_id = as.character(p_id)) %>%
      select(p_id, visual_verbal_score),
    by = "p_id"
  )
```

Model 2 with leraner's type

```
model2 <- glmer(
  RecallAcc ~ TaskType * doodle_type * visual_verbal_score + (1 | p_id),
```

```
  family = binomial,
  data = long_df
)
```

```
## Warning in checkConv(attr(opt, "derivs"), opt$par, ctrl = control$checkConv, : Model failed to conver
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```
print(model2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: RecallAcc ~ TaskType * doodle_type * visual_verbal_score + (1 |
##     p_id)
##    Data: long_df
##       AIC       BIC    logLik -2*log(L)  df.resid
## 1574.6783 1641.8197 -774.3392 1548.6783      1280
## Random effects:
##  Groups Name        Std.Dev.
##  p_id   (Intercept) 0.3366
## Number of obs: 1293, groups:  p_id, 54
## Fixed Effects:
##                                                 (Intercept)
##                                                     0.96749
##                                              TaskTypeVisual
##                                                     0.38499
##                                             doodle_typeFree
##                                                    -0.45701
##                                       doodle_typeStructured
##                                                    -0.11907
##                                         visual_verbal_score
##                                                     0.04635
##                             TaskTypeVisual:doodle_typeFree
##                                                     0.22458
##                       TaskTypeVisual:doodle_typeStructured
##                                                    -0.26315
##                         TaskTypeVisual:visual_verbal_score
##                                                    -0.02732
##                        doodle_typeFree:visual_verbal_score
##                                                    -0.06024
##                  doodle_typeStructured:visual_verbal_score
##                                                    -0.01173
##       TaskTypeVisual:doodle_typeFree:visual_verbal_score
##                                                     0.01277
## TaskTypeVisual:doodle_typeStructured:visual_verbal_score
##                                                     0.01819
## optimizer (Nelder_Mead) convergence code: 0 (OK) ; 0 optimizer warnings; 1 lme4 warnings
```

```
summary(model2)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
```

```
##  Family: binomial  ( logit )
## Formula: RecallAcc ~ TaskType * doodle_type * visual_verbal_score + (1 |
##     p_id)
##    Data: long_df
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1574.7    1641.8    -774.3    1548.7      1280
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2773 -1.2610  0.5519  0.6801  0.9990
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  p_id   (Intercept) 0.1133   0.3366
## Number of obs: 1293, groups:  p_id, 54
##
## Fixed effects:
##                                                            Estimate Std. Error
## (Intercept)                                                 0.96749    0.25295
## TaskTypeVisual                                              0.38499    0.35805
## doodle_typeFree                                            -0.45701    0.32855
## doodle_typeStructured                                      -0.11907    0.33784
## visual_verbal_score                                         0.04635    0.03877
## TaskTypeVisual:doodle_typeFree                              0.22458    0.48727
## TaskTypeVisual:doodle_typeStructured                       -0.26315    0.49147
## TaskTypeVisual:visual_verbal_score                         -0.02732    0.05497
## doodle_typeFree:visual_verbal_score                        -0.06024    0.05087
## doodle_typeStructured:visual_verbal_score                  -0.01173    0.05191
## TaskTypeVisual:doodle_typeFree:visual_verbal_score          0.01277    0.07577
## TaskTypeVisual:doodle_typeStructured:visual_verbal_score    0.01819    0.07560
##                                                            z value Pr(>|z|)
## (Intercept)                                                  3.825 0.000131 ***
## TaskTypeVisual                                               1.075 0.282264
## doodle_typeFree                                             -1.391 0.164228
## doodle_typeStructured                                       -0.352 0.724509
## visual_verbal_score                                          1.195 0.231903
## TaskTypeVisual:doodle_typeFree                               0.461 0.644874
## TaskTypeVisual:doodle_typeStructured                        -0.535 0.592352
## TaskTypeVisual:visual_verbal_score                          -0.497 0.619235
## doodle_typeFree:visual_verbal_score                         -1.184 0.236313
## doodle_typeStructured:visual_verbal_score                   -0.226 0.821265
## TaskTypeVisual:doodle_typeFree:visual_verbal_score           0.168 0.866208
## TaskTypeVisual:doodle_typeStructured:visual_verbal_score     0.241 0.809899
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) TskTyV ddl_tF ddl_tS vsl_v_ TsTV:_F TsTV:_S TTV:__ d_F:__
## TaskTypeVsl -0.647
## doodl_typFr -0.707  0.498
## ddl_typStrc -0.687  0.485  0.529
## vsl_vrbl_sc  0.792 -0.514 -0.561 -0.545
## TskTypVs:_F  0.476 -0.735 -0.674 -0.356  0.378
```

```
## TskTypVs:_S  0.472 -0.728 -0.363 -0.687  0.375  0.535
## TskTypVs:__ -0.514  0.788  0.395  0.384 -0.646 -0.579  -0.574
## ddl_typF:__ -0.556  0.392  0.780  0.416 -0.698 -0.525  -0.286   0.492
## ddl_typS:__ -0.544  0.384  0.419  0.789 -0.684 -0.282  -0.543   0.482  0.521
## TskTV:_F:__  0.373 -0.572 -0.523 -0.279  0.469  0.771   0.417  -0.725 -0.671
## TskTV:_S:__  0.373 -0.573 -0.287 -0.542  0.470  0.421   0.786  -0.727 -0.358
##             d_S:__ TTV:_F:
## TaskTypeVsl
## doodl_typFr
## ddl_typStrc
## vsl_vrbl_sc
## TskTypVs:_F
## TskTypVs:_S
## TskTypVs:__
## ddl_typF:__
## ddl_typS:__
## TskTV:_F:__ -0.350
## TskTV:_S:__ -0.687  0.528
## optimizer (Nelder_Mead) convergence code: 0 (OK)
## Model failed to converge with max|grad| = 0.0120035 (tol = 0.002, component 1)
##   See ?lme4::convergence and ?lme4::troubleshooting.
```

```
long_df <- long_df %>%
  mutate(
    visual_verbal_c = scale(visual_verbal_score, center = TRUE, scale = FALSE)[,1]
  )

model3 <- glmer(
  RecallAcc ~ TaskType * doodle_type * visual_verbal_c + (1 | p_id),
  family = binomial,
  data = long_df,
  control = glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e5))
)

vv_sd <- sd(long_df$visual_verbal_c, na.rm = TRUE)

vv_vals <- c(-vv_sd, 0, vv_sd)
vv_vals
```

```
## [1] -4.092454  0.000000  4.092454
```

```
emm3 <- emmeans(
  model3,
  ~ doodle_type * TaskType | visual_verbal_c,
  at = list(visual_verbal_c = vv_vals),
  type = "response"
)

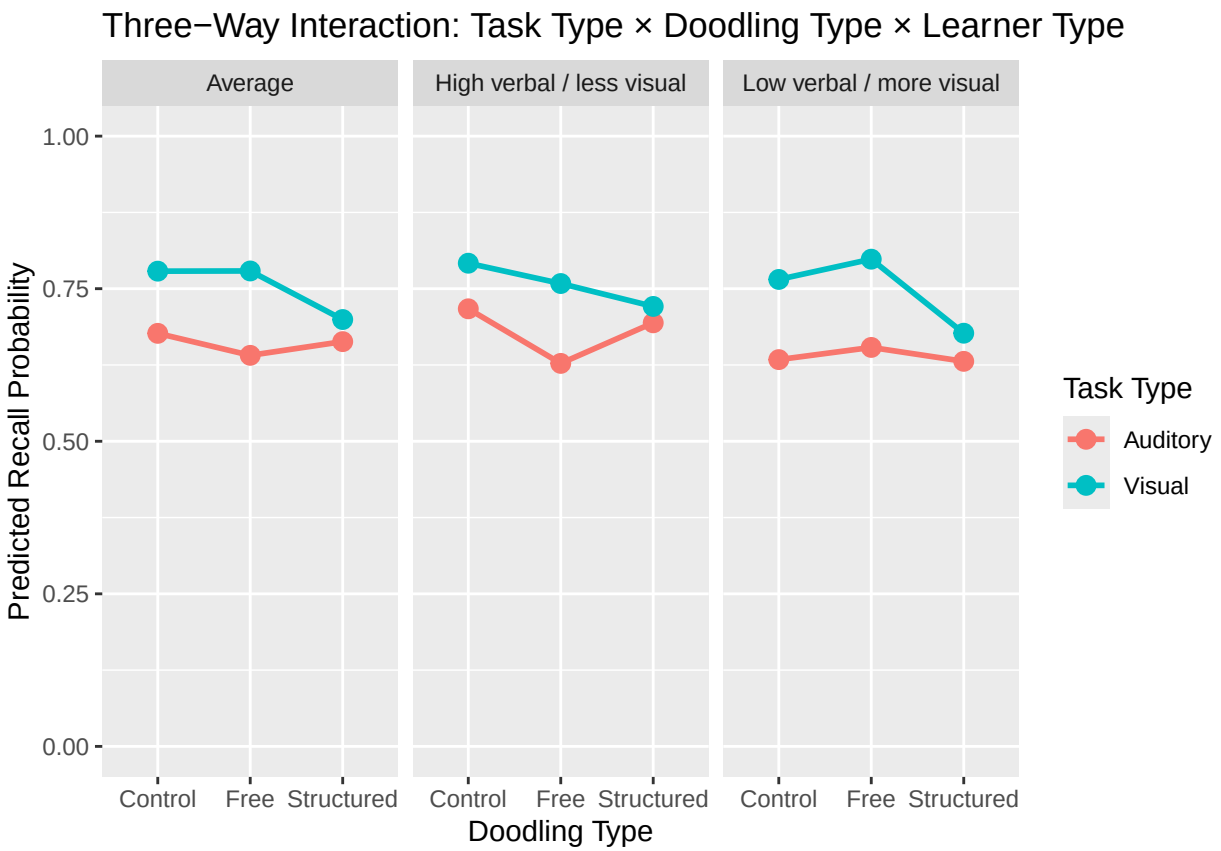
emm3_df <- as.data.frame(emm3)
emm3_df
```

```
## visual_verbal_c = -4.092454:
##  doodle_type TaskType      prob         SE  df asymp.LCL asymp.UCL
```

```
##  Control     Auditory 0.6337995 0.04983141 Inf 0.5318934 0.7249911
##  Free        Auditory 0.6537628 0.04844710 Inf 0.5538232 0.7417551
##  Structured  Auditory 0.6309843 0.04995816 Inf 0.5289470 0.7225140
##  Control     Visual   0.7651731 0.04279081 Inf 0.6713938 0.8386220
##  Free        Visual   0.7983788 0.03935354 Inf 0.7103379 0.8647551
##  Structured  Visual   0.6770737 0.04787406 Inf 0.5771796 0.7630557
##
## visual_verbal_c =  0.000000:
##  doodle_type TaskType      prob         SE  df asymp.LCL asymp.UCL
##  Control     Auditory 0.6768313 0.03382242 Inf 0.6073478 0.7392969
##  Free        Auditory 0.6407935 0.03475274 Inf 0.5702545 0.7057291
##  Structured  Auditory 0.6632718 0.03430324 Inf 0.5931197 0.7268979
##  Control     Visual   0.7787496 0.02947407 Inf 0.7156765 0.8311323
##  Free        Visual   0.7790772 0.02948508 Inf 0.7159656 0.8314666
##  Structured  Visual   0.6994692 0.03297154 Inf 0.6311965 0.7599106
##
## visual_verbal_c =  4.092454:
##  doodle_type TaskType      prob         SE  df asymp.LCL asymp.UCL
##  Control     Auditory 0.7170638 0.04631427 Inf 0.6183433 0.7985674
##  Free        Auditory 0.6276152 0.04931577 Inf 0.5270810 0.7182037
##  Structured  Auditory 0.6941038 0.04725820 Inf 0.5946288 0.7782707
##  Control     Visual   0.7917548 0.04077688 Inf 0.7007369 0.8605975
##  Free        Visual   0.7584867 0.04247634 Inf 0.6659513 0.8318607
##  Structured  Visual   0.7209516 0.04568250 Inf 0.6234302 0.8012684
##
## Confidence level used: 0.95
## Intervals are back-transformed from the logit scale
```

```r
emm3_df <- emm3_df %>%
  mutate(
    learner_group = case_when(
      visual_verbal_c == vv_vals[1] ~ "Low verbal / more visual",
      visual_verbal_c == vv_vals[2] ~ "Average",
      visual_verbal_c == vv_vals[3] ~ "High verbal / less visual"
    )
  )

ggplot(emm3_df, aes(x = doodle_type, y = prob, color = TaskType, group = TaskType)) +
  geom_point(size = 3) +
  geom_line(linewidth = 1) +
  facet_wrap(~ learner_group) +
  labs(
    x = "Doodling Type",
    y = "Predicted Recall Probability",
    color = "Task Type",
    title = "Three-Way Interaction: Task Type × Doodling Type × Learner Type"
  ) +
  coord_cartesian(ylim = c(0, 1))
```



MPI

```r
# reverse code
mpi_items <- paste0("MPI_", 1:14)

reverse_items <- c("MPI_5", "MPI_6", "MPI_8", "MPI_10",
                   "MPI_11", "MPI_13", "MPI_14")

df <- df %>%
  mutate(
    across(all_of(reverse_items), ~ 6 - .)
  )
```

```r
# sum
df <- df %>%
  rowwise() %>%
  mutate(
    MPI_score = sum(c_across(all_of(mpi_items)), na.rm = TRUE)
  ) %>%
  ungroup()

# merge to long dataset
long_df <- long_df %>%
```

```
  mutate(p_id = as.character(p_id)) %>%
  left_join(
    df %>%
      mutate(p_id = as.character(p_id)) %>%
      select(p_id, MPI_score),
    by = "p_id"
  )
```

```
# center the MPI score
long_df <- long_df %>%
  mutate(
    MPI_c = scale(MPI_score, center = TRUE, scale = FALSE)[,1]
  )
```

```
# model with MPI score
model_mpi <- glmer(
  RecallAcc ~ TaskType * doodle_type * MPI_c + (1 | p_id),
  family = binomial,
  data = long_df
)

summary(model_mpi)
```

```
## Generalized linear mixed model fit by maximum likelihood (Laplace
##   Approximation) [glmerMod]
##  Family: binomial  ( logit )
## Formula: RecallAcc ~ TaskType * doodle_type * MPI_c + (1 | p_id)
##    Data: long_df
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##    1574.3    1641.4    -774.1    1548.3      1280
##
## Scaled residuals:
##     Min      1Q  Median      3Q     Max
## -2.2452 -1.2459  0.5634  0.6700  0.9992
##
## Random effects:
##  Groups Name        Variance Std.Dev.
##  p_id   (Intercept) 0.1176   0.3429
## Number of obs: 1293, groups:  p_id, 54
##
## Fixed effects:
##                                          Estimate Std. Error z value
## (Intercept)                              0.734281   0.154173   4.763
## TaskTypeVisual                           0.524691   0.220078   2.384
## doodle_typeFree                         -0.155164   0.205304  -0.756
## doodle_typeStructured                   -0.051405   0.207522  -0.248
## MPI_c                                    0.004003   0.016695   0.240
## TaskTypeVisual:doodle_typeFree           0.162041   0.310279   0.522
## TaskTypeVisual:doodle_typeStructured    -0.360593   0.304035  -1.186
## TaskTypeVisual:MPI_c                     0.002319   0.024052   0.096
## doodle_typeFree:MPI_c                    0.001594   0.022271   0.072
## doodle_typeStructured:MPI_c             -0.025049   0.022232  -1.127
```

```
## TaskTypeVisual:doodle_typeFree:MPI_c        0.010500   0.034175   0.307
## TaskTypeVisual:doodle_typeStructured:MPI_c  0.003893   0.032736   0.119
##                                            Pr(>|z|)
## (Intercept)                                1.91e-06 ***
## TaskTypeVisual                               0.0171 *
## doodle_typeFree                              0.4498
## doodle_typeStructured                        0.8044
## MPI_c                                        0.8105
## TaskTypeVisual:doodle_typeFree               0.6015
## TaskTypeVisual:doodle_typeStructured         0.2356
## TaskTypeVisual:MPI_c                         0.9232
## doodle_typeFree:MPI_c                        0.9429
## doodle_typeStructured:MPI_c                  0.2599
## TaskTypeVisual:doodle_typeFree:MPI_c         0.7587
## TaskTypeVisual:doodle_typeStructured:MPI_c   0.9053
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##             (Intr) TskTyV ddl_tF ddl_tS MPI_c  TsTV:_F TsTV:_S TTV:MP d_F:MP
## TaskTypeVsl -0.632
## doodl_typFr -0.680  0.475
## ddl_typStrc -0.672  0.471  0.505
## MPI_c        0.013 -0.008 -0.009 -0.009
## TskTypVs:_F  0.450 -0.709 -0.662 -0.334  0.006
## TskTypVs:_S  0.458 -0.724 -0.344 -0.682  0.006  0.513
## TskTyV:MPI_ -0.008  0.024  0.006  0.006 -0.631 -0.017  -0.017
## ddl_tF:MPI_ -0.009  0.006  0.013  0.007 -0.682 -0.009  -0.005   0.473
## ddl_tS:MPI_ -0.010  0.006  0.007 -0.023 -0.683 -0.005   0.016   0.474  0.512
## TTV:_F:MPI_  0.006 -0.017 -0.009 -0.004  0.444  0.042   0.012  -0.704 -0.652
## TTV:_S:MPI_  0.006 -0.018 -0.005  0.016  0.464  0.012  -0.013  -0.735 -0.348
##             d_S:MP TTV:_F:
## TaskTypeVsl
## doodl_typFr
## ddl_typStrc
## MPI_c
## TskTypVs:_F
## TskTypVs:_S
## TskTyV:MPI_
## ddl_tF:MPI_
## ddl_tS:MPI_
## TTV:_F:MPI_ -0.334
## TTV:_S:MPI_ -0.679  0.517
```

```r
# get prediction
mpi_range <- seq(min(long_df$MPI_c), max(long_df$MPI_c), length.out = 50)

emm_mpi <- emmeans(
  model_mpi,
  ~ doodle_type * TaskType | MPI_c,
  at = list(MPI_c = mpi_range),
  type = "response"
)
```

```r
emm_mpi_df <- as.data.frame(emm_mpi)
```

```r
ggplot(emm_mpi_df, aes(x = MPI_c, y = prob, color = doodle_type)) +
  geom_line(size = 1) +
  facet_wrap(~ TaskType) +
  labs(
    x = "Multitasking Preference (MPI Centered)",
    y = "Predicted Recall Probability",
    color = "Doodling Type",
    title = "Model-Predicted Interaction: MPI × Doodling × Task"
  ) +
  coord_cartesian(ylim = c(0, 1))
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

